AIMERS CLASSES

SAMPLE QUESTION PAPER

SESSION 2025-2026

Class 11th

PHYSICS

SET-02

Hints & Solutions

1. (A)

(i) (A) velocity has dimensions LT^{-1}.

(ii) (C) constant acceleration $\Rightarrow$ linear $v-t$ relation.

(iii) (C) $\frac{180}{\pi} \approx 57.3^\circ$

(iv) (B) force is derived $\left(MLT^{-2}\right)$.

(v) (A) slope $\frac{ds}{dt} = v$.

(vi) (A) potential energy per unit mass.

(vii) (B) momentum conserved; KE not conserved.

1. (B)

(i) Rate of change of angular displacement; $\omega = \frac{d\theta}{dt}$. SI unit: rad s^{-1}.

(ii) Newton's Second Law of Motion states that force (F) is directly proportional to mass (m) and acceleration (a). This relationship is given by the equation: F = ma

(iii) 90° (velocity horizontal, acceleration vertical).

(iv) N•m.

(v) Work and heat (both joule).

(vi) $KE = p^2/(2m) \Rightarrow$ becomes 4 times of KE.

(vii) $KE = \frac{1}{2}mv^2 = \frac{1}{2}\times 2\times 9 = 9\ J$.

2. (a) Centripetal acceleration $a_c = v^2/r$. From Newton II:

$$F_c = ma_c = m\frac{v^2}{r}(\text{ towards centre})$$

OR

The line joining a planet and the Sun sweeps equal areas in equal times (areal velocity constant). It follows from conservation of angular momentum under a central force:

3.

- Cannot give numerical constants or dimensionless factors (like $2, \pi$).
- Cannot distinguish between physical quantities with same dimensions (e.g., torque and work).
- Cannot be used if more than one quantity of same dimension is involved in an equation.

4. $a = \frac{F}{m} = \frac{40}{5} = 8\ \text{m s}^{-2},\ u = 0$

$v = u + at = 0 + 8 \times 4 = 32\ \text{m s}^{-1}$

$s = ut + \frac{1}{2}at^2 = 0 + 0.5 \times 8 \times 16 = 64\ \text{m}$

$v = 32\ \text{m s}^{-1}, s = 64\ \text{m}$

OR

$m = 0.02\ \text{kg}, a = \frac{\Delta v}{\Delta t} = \frac{0 - 200}{0.02} = -10{,}000\ \text{m s}^{-2}.$

$F = ma = 0.02 \times (-10{,}000) = -200\ \text{N}$ (opposite to motion).

200 N (retarding).

5. (head-on elastic): $v_1' = \frac{m_1 - m_2}{m_1 + m_2} u_1$

$= \frac{2-3}{5} \times 10 = -2\ \text{m s}^{-1}$

$v_2' = \frac{2m_1}{m_1 + m_2} u_1 = \frac{4}{5} \times 10 = 8\ \text{m s}^{-1}$

$v_1' = -2\ \text{m s}^{-1}$ (reverses), $v_2' = 8\ \text{m s}^{-1}$.

6. $k = \frac{F}{x} = \frac{10}{0.05} = 200\ \text{N m}^{-1}$

Work to reach $x = 0.10\ \text{m}: W = \frac{1}{2}kx^2$

$= 0.5 \times 200 \times 0.01 = 1.0\ \text{J}.$

= 1.0 J.

7. $a = \frac{v - u}{t} = 2\ \text{m s}^{-2}$

$s = ut + \frac{1}{2}at^2 = 0 + 0.5 \times 2 \times 100 = 100\ \text{m}$

$a = 2\ \text{m s}^{-2}, s = 100\ \text{m}$

8. (any two with brief note): gun recoil, rocket propulsion, billiard collisions, explosions, hose recoil.

9. $u_x = 50\cos 30° = 43.3, u_y = 50\sin 30° = 25.$

$T = \frac{2u\sin\theta}{g} = \frac{2 \times 50 \times 0.5}{9.8} = 5.10\ \text{s}$

$H = \frac{u^2 \sin^2\theta}{2g} = \frac{2500 \times 0.25}{19.6} = 31.9\ \text{m}$

$R = \frac{u^2 \sin 2\theta}{g} = \frac{2500 \times \sin 60°}{9.8}$

$= \frac{2500 \times 0.866}{9.8} \approx 221\ \text{m}$

$T \approx 5.10\ \text{s}, H \approx 31.9\ \text{m}, R \approx 221\ \text{m}$

OR

Max tension $T_{max} = 72\text{kgf} = 72g\ \text{N}$. Man's weight $mg = 96g\ \text{N}$.

For descending with acceleration a : upward force T balances $m(g-a) \Rightarrow T = m(g-a)$.

Max safe $T = T_{max}: m(g-a) = 72g$

$\Rightarrow 96g - 96a = 72g$

$\Rightarrow 96a = 24g$

$\Rightarrow a = \frac{24}{96}g = \frac{g}{4} \approx 2.45\ \text{m/s}^2$ downward.

10. $\vec{A} \cdot \vec{B} = 2(1) + 1(-1) + 2(2) = 5, |A| = 3, |B| = \sqrt{6}$

$\cos\theta = \frac{5}{3\sqrt{6}} \Rightarrow \theta = \cos^{-1}\left(\frac{5}{3\sqrt{6}}\right) \approx 47.8°$

$\theta \approx 47.8°$

11. Distance in n^{th} second (from rest): $s_n = \frac{g}{2}(2n-1) = g\left(n - \frac{1}{2}\right).$

$9.8(n - 0.5) = 25 \Rightarrow n = 3.051\ \text{s}.$

Total height $H = \frac{1}{2}gn^2 = 4.9 \times (3.051)^2 \approx 45.6\ \text{m}.$

$H \approx 46\ \text{m}$ (approx.)

12. $\omega = \sqrt{k/m} = 20\ \text{rads}^{-1};$

$f = \omega / 2\pi = 20/(2\pi) = 3.18\ \text{Hz}$

$f \approx 3.18\ \text{Hz}$

[2]

13. $f_2 = 320 \pm 8 \Rightarrow 312$ Hz or 328 Hz.
312 Hz or 328 Hz.

14. Let each half be d.

$t_1 = \frac{d}{60}, t_2 = \frac{d}{90}$

$\Rightarrow t = \frac{d}{60} + \frac{d}{90} = \frac{d}{36}$

Total distance $2d$. Avg speed

$= \frac{2d}{t} = \frac{2d}{d/36} = 72 \text{ km h}^{-1}$.

72 km / h

15. Take the direction of the incoming velocity as positive.
Given: $m = 0.50 \text{ kg}, u = +6.0 \text{ m/s}, v = -4.0 \text{ m/s}$ (opposite direction), $\Delta t = 0.020 \text{ s}$.

1. Impulse $J = \Delta p = m(v - u)$.

Compute step by step:
$v - u = -4.0 - 6.0 = -10.0 \text{ m/s}$.
$m(v - u) = 0.50 \times (-10.0) = -5.0 \text{ N·s}$.

So, $J = -5.0 \text{ N·s}$. (Magnitude $= 5.0 \text{ N·s}$.
The negative sign shows the impulse acts opposite to the original direction - it pushes the ball away from the wall.)

2. Average force $F_{avg} = \frac{J}{\Delta t} = \frac{-5.0}{0.020} = -250 \text{ N}$.

So, the average force on the ball is -250 N (magnitude 250 N), directed away from the wall.

OR

$\Rightarrow f_g = mg, F_a = ma$

$F_{totla} = F_g + F_a$

16. Let $|\vec{A}| = |\vec{B}| = A$.

$|\vec{R}| = \sqrt{A^2 + A^2 + 2A^2\cos\theta} = A \Rightarrow 2(1 + \cos\theta) = 1.$

$\Rightarrow \cos\theta = -\frac{1}{2} \Rightarrow \theta = 120°.$

17. Fundamental frequencies:

- Open pipe: $f_o = \frac{v}{2L} = \frac{330}{2 \times 0.30} = 550 \text{ Hz}$.
- Closed pipe: $f_c = \frac{v}{4L} = \frac{330}{4 \times 0.30} = 275 \text{ Hz}$.

Beat frequency initially $= |550 - 275| = 275$ Hz
(given 5 in prompt is just the condition motivating adjustment).
To match frequencies, adjust closed pipe to have $f_c' = f_o = 550$ Hz :
For closed pipe,
$f_c' = \frac{v}{4L'} = 550 \Rightarrow L' = \frac{v}{4 \times 550} = \frac{330}{2200} = 0.15 \text{ m} =$ 15 cm.
So change in length $= |L' - L| = |15 - 30| = 15$ cm
(shorten by 15 cm).

18. Let $f = kr^a \rho^b \sigma^c$.
Dimensions:

$[f] = T^{-1}$
$[r] = L$
$[\rho] = ML^{-3}$
$[\sigma] = MT^{-2}$
So:

$T^{-1} = L^a (ML^{-3})^b (MT^{-2})^c = M^{b+c} L^{a-3b} T^{-2c}$

Equating powers:

- For $M: b + c = 0 \Rightarrow b = -c$
- For $L: a - 3b = 0 \Rightarrow a = 3b$
- For $T: -2c = -1 \Rightarrow c = 0.5$

Then $b = -0.5, a = -1.5$.

$$f \propto \frac{\sigma^{1/2}}{\rho^{1/2} r^{3/2}}$$

$$f \propto \sqrt{\frac{\sigma}{\rho r^3}}$$

OR

Speed of source $v_s = \frac{2\pi r}{T} = \frac{2\pi \times 15}{3} = 31.4 \text{ m/s}$.

Actual frequency $f = 200$ Hz.

- Max frequency:

$$f_{max} = \frac{v}{v - v_s} f = \frac{330}{330 - 31.4} \times 200 = 218.9 \text{ Hz}.$$

- Min frequency:

$$f_{min} = \frac{v}{v + v_s} f = \frac{330}{330 + 31.4} \times 200 = 182.9 \text{ Hz}.$$

$f_{max} \approx 219$ Hz, $f_{min} \approx 183$ Hz.

19. COP of refrigerator $= \frac{T_L}{T_H - T_L}$.

$T_L = 270\text{K}, T_H = 300\text{K}$.

$$\text{COP} = \frac{270}{30} = 9$$

Work $W = 1$ J.

Heat absorbed from cold space

$Q_L = \text{COP} \times W = 9$ J.

Heat delivered to room:

$Q_H = Q_L + W = 9 + 1 = 10$ J.

10 J of heat delivered per joule of electrical work.

OR

Components:

$u_x = 10\cos 30° = 8.66 \text{ m/s}, u_y = 10\sin 30° = 5 \text{ m/s}.$

Horizontal range: $x = u_x t = 8.66t = 17.3$.

So $t = 2$ s.

Vertical displacement:

$$y = u_y t - \frac{1}{2} g t^2 = 5(2) - 5(4) = 10 - 20 = -10 \text{ m}.$$

So height of tower $= 10$ m.

20.

1. From definition: $a = \frac{dv}{dt}$. Integrate:

$$\int dv = \int a\,dt \Rightarrow v = u + at$$

2. Displacement: $v = \frac{ds}{dt} \Rightarrow \int ds = \int (u + at)\,dt$.

$$s = ut + \frac{1}{2} a t^2$$

3. Eliminate t : From 1, $t = \frac{v - u}{a}$. Sub in 2 :

$$s = \frac{(v + u)}{2} \cdot \frac{v - u}{a}$$

$$\Rightarrow v^2 = u^2 + 2as$$

Equations derived are:

$$v = u + at$$

$$s = ut + \frac{1}{2} a t^2$$

$$v^2 = u^2 + 2as$$

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SAMPLE QUESTION PAPER

SESSION 2025-2026

PHYSICS

Class 11th

SET-02

Time: 120 Minutes

M.M.: 70

General Instructions:

1. Answers to this Paper must be written on the paper provided separately.
2. You will not be allowed to write during first 15 minutes.
3. This time is to be spent in reading the question paper.
4. The time given at the head of this Paper is the time allowed for writing theanswers.
5. Section A is compulsory.
6. The intended marks for questions or parts of questions are given in brackets [ ].

SECTION-A (14 Marks)

(Attempt all questions from this Section.)

1. In questions (i) to (vii) below, choose the correct alternative (a), (b), (c), or (d) for each of the questions given below:

(i) The dimensional formula of velocity is:

(A) $\left[M^0\ L^1\ T^{-1}\right]$

(B) $\left[M^1\ L^1\ T^{-1}\right]$

(C) $\left[M^0\ L^1\ T^0\right]$

(D) $\left[M^{-1}\ L^1\ T^{-2}\right]$

(ii) If a body is uniformly accelerated, its $v-t$ graph is:

(A) A curve

(B) A parabola

(C) A straight line

(D) A horizontal line

(iii) One radian ≈

(A) 45°

(B) 90°

(C) 57.3°

(D) 180°

(iv) Which is not fundamental?

(A) Mass

(B) Force

(C) Length

(D) Time

(v) Slope of a displacement-time graph gives:

(A) Velocity

(B) Acceleration

(C) Jerk

(D) Momentum

(vi) Unit of gravitational potential is:

(A) Jkg^{-1}

(B) J

(C) $N \cdot m$

(D) $m^2\ s^{-2}$

(vii) In a perfectly inelastic collision:

(A) Both momentum and kinetic energy conserved

(B) Momentum conserved but KE not conserved

(C) KE conserved but momentum not conserved

(D) Neither conserved

(B) Answer the following questions briefly:

(i) Define angular velocity.

(ii) What is the relationship between force, mass, and acceleration as described by Newton's Second Law of Motion?

(iii) Angle between velocity and acceleration at the highest point of projectile motion?

(iv) SI unit of torque.

(v) Two scalar quantities with same dimensions as energy.

(vi) If momentum doubles, what happens to kinetic energy?

(vii) A body of mass 2 kg has velocity $3\ m\ s^{-1}$. Find its KE.

SECTION B – 14 MARKS

2. Derive centripetal force on a body moving with speed v in a circle of radius r.

OR

State Kepler's second law and explain.

3. State any two limitations of dimensional analysis.

4. A 40 N force acts on a 5 kg body at rest for 4 s. Find final velocity and displacement.

OR

A 20 g bullet at $200\ m/s$ stops in 0.02 s inside a target. Find the average retarding force.

5. A 2 kg body moving at $10\ m/s$ collides elastically with a stationary 3 kg body. Find final velocities.

6. A spring extends 5 cm under 10 N. Find the work to stretch it to 10 cm.

7. A car accelerates uniformly from rest to $20\ m/s$ in 10 s. Find acceleration and distance.

8. State two applications of conservation of linear momentum.

SECTION C – 27 MARKS

9. A projectile is fired with $u = 50\ m\ s^{-1}$ at $30°$. Find (i) H_{max}, (ii) Range, (iii) Time of flight ($g = 9.8$).

OR

A string can bear a maximum tension of 72 kgf. With what maximum downward acceleration can a man of mass $m = 96,000\ g$ descend using this rope?

10. Find the angle between $\vec{A} = 2\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{B} = \hat{i} - \hat{j} + 2\hat{k}$.

11. A body dropped from rest covers 25 m in the last second of its fall. Find the height of fall $(g = 9.8)$.

12. A spring has $k = 400\ N\ m^{-1}$. A mass $m = 1.0\ kg$ is attached. Find frequency of SHM.

13. Two tuning forks produce 8 beats s^{-1}. One fork has 320 Hz. Find possible frequencies of the other.

14. A car travels half the distance at $60\ km/h$ and the other half at $90\ km/h$. Find average speed.

15. A ball of mass $m = 0.50\ kg$ is moving with velocity $u = 6.0\ m\,s^{-1}$.
It strikes a smooth vertical wall normally and rebounds with velocity $v = 4.0\ m\,s^{-1}$ in the opposite direction.
Calculate the impulse imparted to the ball and the average force on the ball if the contact time is $\Delta t = 0.020\ s$.

OR

A rocket with a lift-off mass of 20,000 kg is blasted upwards with an initial acceleration of 5.0 m/sec square. Calculate the initial thrust (force) of the blast.

16. Find the angle between two equal vectors such that the magnitude of the resultant equals either vector.

17. A closed pipe and an open pipe sound together and produce 5 beats per second. If the common length is $L = 30\ cm$, by how much should the length of the closed pipe be changed so that they produce the same frequency? Take $v = 330\ m/s$.

SECTION D – 15 MARKS

18. The frequency f of oscillation of a liquid drop depends on its radius (r), density (ρ), and surface tension (σ). Use dimensional analysis to obtain the formula for f.

OR

A source of sound moves in a circle of radius 15 m with frequency 200 Hz, completing one revolution in 3 s. A listener is 30 m away from the center. Find the maximum and minimum frequency heard. ($v = 330\ m/s$)

19. The temperature inside a refrigerator is $-3°C$ and the room temperature is $27°C$. How many joules of heat will be delivered to the room for each joule of electrical energy consumed ideally?

OR

A ball is thrown from the top of a tower with initial velocity $10\ m/s$ at angle 30° above the horizontal. It hits the ground 17.3 m away from the base. Find the height of the tower. ($g = 10\ m/s^2$)

20. Derive the three equations of motion using the calculus method.

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